

Tiago Davies

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WORK EXPERIENCE

Software Engineer

Aug 2025 – Present

Smart Software Solutions

- Expanded a cross-platform **.NET RESTful API** to support secure **JWT authentication** using a refresh token system for a mobile application while maintaining cookie-based sessions for the web platform.
- Enabled seamless user authentication across iOS and Android by integrating a **Cordova** biometrics plugin, allowing for secure Face ID and fingerprint logins.
- Modernized database transaction handling by introducing rollback/commit patterns and structured error handling, improving reliability and reducing data integrity issues.
- Refactored **Angular front-end** architecture from template driven to reactive forms, enforcing consistent client side validation and error handling.
- Integrated Tesseract Optical Character Recognition to extract and auto-fill fields with Machine Readable Zone data from passports, achieving over 90% parsing accuracy.
- Transformed Figma designs into interactive and **responsive front-end views**, using modern CSS and Bootstrap to achieve fluid UI across multiple screen sizes on both web and mobile.
- Developed data-driven admin customization pages empowering clients to self-manage pages, email contents, and dynamic field layouts.
- Designed and implemented a **relational database schema** and business logic for a multi-allocation payment system, allowing flexible partial payments against outstanding balances.
- Automated build deployments to the development server via bash scripts, reducing build times by 30%.

Computer Science Teaching Assistant

Jan 2023 – May 2024

University of Rochester

- Spearheaded weekly technical labs for 150+ students in core Data Structures & Algorithms topics.
- Guided students through problem-solving, debugging, and code architecture during study sessions.

Computer Science Workshop Leader

Jan 2022 – Dec 2022

University of Rochester

- Led weekly workshops for groups of 10+ students on data structures and algorithm topics.
- Participated in weekly leadership class focused on teamwork and communication.

SKILLS

- **Languages:** C#, TypeScript, JavaScript, T-SQL
- **Back-end:** .NET 8/.NET (Core), Entity Framework, LINQ, ASP.NET Web API
- **Front-end:** Angular 17, React, HTML, CSS/SCSS, Bootstrap, jQuery
- **Database:** Microsoft SQL Server
- **Tools:** Docker, Git/GitHub, npm, Vite, Cordova, TortoiseSVN, Postman, VS Code, Visual Studio

PROJECTS

Entity Component System Web Game (Rogue Survivor)

[May 2026](#)

- Engineered a high performance data-oriented web game in TypeScript by leveraging an Entity Component System, optimizing for cache locality via flat TypedArrays to handle 2,000+ active entities.
- Built a custom spatial hashing collision system, and sprite pooling to reduce garbage collection.

EDUCATION

University of Rochester

Graduated May 2025

Bachelor of Science, Computer Science

- Cumulative GPA: 3.81 (out of 4.0)
- Tom Hope Memorial Media Scholar Award
- Water Polo Club Business Manager